How likely is it that the TRM rises by more than 30 pesos in one day?

A normal model for daily changes in Colombia's TRM (C10)

Valeria Barrios Forero — Grade 11, 11A — Global Economics 2026–2027

1 Introduction and methodology

Question. For an importer who pays in dollars, a sharp jump in the Tasa Representativa del Mercado (TRM, the official COP/USD exchange rate) raises costs from one day to the next. I ask: *how likely is it that the TRM increases by more than 30 COP/USD between two consecutive certifications?*

Variable. X = change in the TRM between two consecutive certifications, in Colombian pesos per US dollar (COP/USD). Target population: daily changes of Colombia's TRM. Observation period: certifications effective from August 1, 2026 to November 5, 2026.

Source. Superintendencia Financiera de Colombia, *Tasa de Cambio Representativa del Mercado – TRM*, Portal Nacional de Datos Abiertos (datos.gov.co), updated November 5, 2026; accessed and downloaded as CSV on November 5, 2026. Link to the original source: datos.gov.co/d/32sa-8pi3. Exact dataset used (dates, TRM and changes): file on Drive/OneDrive

Data preparation. The full CSV has 8,382 certifications (1991–2026). I selected the 64 TRM values effective from 08/01/2026 to 11/05/2026 (a short, recent period so conditions are comparable). There are no missing values or repeated dates, and I removed no observations. The certification effective 11/06/2026 was excluded because it starts after my access date. Transformation: difference between consecutive TRM values, $x_i = \text{TRM}_i - \text{TRM}_{i-1}$ (COP/USD), which gives $n = 63$ observations. Note: Friday's TRM also applies on Saturday and Sunday, so some changes span several calendar days.

2 Results and discussion

2.1 Sample statistics

With $\sum x_i = 77.05$ COP/USD and $n = 63$:

$$\bar{x} = \frac{1}{n}\sum x_i = \frac{77.05}{63} = 1.2230 \text{ COP/USD}.$$

With $\sum (x_i - \bar{x})^2 = 52\,722.82$ (COP/USD)^2:

$$s^2 = \frac{1}{n-1}\sum (x_i - \bar{x})^2 = \frac{52\,722.82}{62} = 850.37 \text{ (COP/USD)}^2, \qquad s = \sqrt{850.37} = 29.16 \text{ COP/USD}.$$

The minimum is -63.71 and the maximum is 86.30 COP/USD.

Classes. I used 9 classes of width 20 COP/USD (close to $\sqrt{63} \approx 8$ classes): wide enough to avoid noise and narrow enough to show the shape, which is unimodal and roughly symmetric with a slightly longer right tail.

2.2 Model and parameters

I use the **normal** model (option C): $\hat{\mu} = \bar{x} = 1.2230$ and $\hat{\sigma} = s = 29.1611$ COP/USD, i.e. $X \sim N(1.2230,\ 850.37)$ with

$$f(x) = \frac{1}{29.1611\sqrt{2\pi}} \exp\left(-\frac{(x - 1.2230)^2}{2(850.37)}\right) \approx 0.01368\, e^{-(x-1.2230)^2/1700.74}, \qquad -\infty < x < \infty.$$

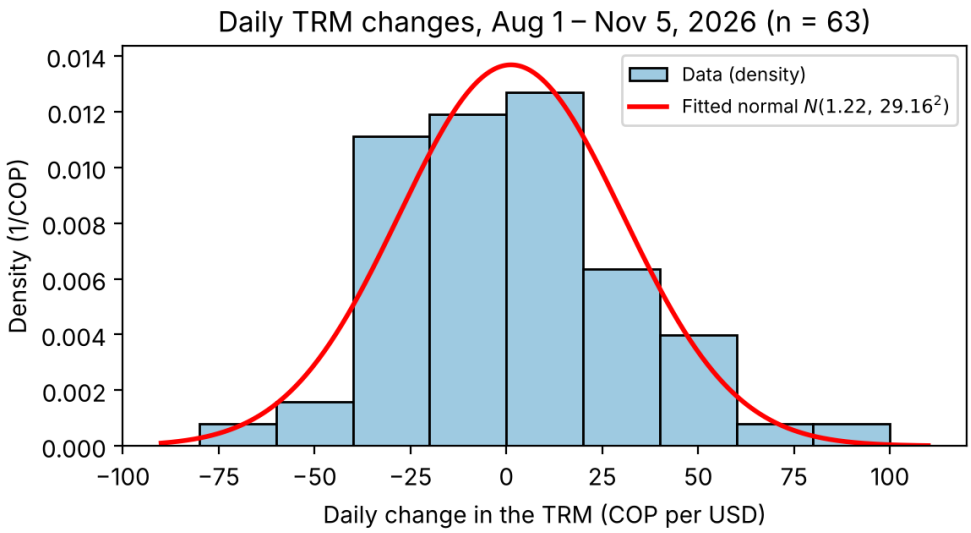


Figure 1: Density histogram with the fitted normal curve. Classes of width 20 COP/USD between −80 and 100 (frequencies: 1, 2, 14, 15, 16, 8, 5, 1, 1).

Justification. The histogram is unimodal, nearly symmetric and has no severe outliers (sample skewness 0.57, excess kurtosis 0.25). Independent shape check: 45 of 63 observations (71.4 %) lie within $\bar{x} \pm s$, versus 68.3 % under the model, and 60 of 63 (95.2 %) lie within $\bar{x} \pm 2s$, versus 95.4 %. The model mean and SD matching $\bar{x}$ and s happens by construction and is not evidence of good fit. **Limitation.** Changes in financial prices often have heavier tails and volatility that clusters over time, and consecutive changes may not be independent; the model also gives positive probability to arbitrarily large changes.

2.3 Quantiles and model probabilities

With $z_{0.10} = -1.28155$ and $z_{0.90} = 1.28155$, using $q_p = \hat{\mu} + \hat{\sigma} z_p$:

$$q_{0.10} = 1.2230 + 29.1611(-1.28155) = 1.2230 - 37.3714 = -36.1484 \text{ COP/USD},$$
$$q_{0.90} = 1.2230 + 29.1611(1.28155) = 1.2230 + 37.3714 = 38.5944 \text{ COP/USD}.$$

By construction: $P(X < q_{0.10}) = 0.10$, $P(q_{0.10} \le X \le q_{0.90}) = 0.80$ and $P(X > q_{0.90}) = 0.10$.

Contextual threshold $c = 30$ COP/USD (a daily rise above 30 pesos noticeably raises the cost of an import; it is not a model quantile):

$$z = \frac{30 - 1.2230}{29.1611} = 0.9868, \qquad P(X > 30) = P(Z > 0.9868) = 1 - 0.8381 = 0.1619.$$

2.4 Empirical vs. model comparison

The same boundaries are used for data and model ($q_{0.10} = -36.1484$, $q_{0.90} = 38.5944$, $c = 30$), counting observations, with $n = 63$. Formulas: $p_{\text{emp}} = \text{count}/n$, $EA = |p_{\text{emp}} - p_{\text{model}}|$, $E\% = 100\, EA/p_{\text{model}}$.

Event	Count	n	p_{emp}	p_{model}	EA	$E\%$
$X < -36.1484$	3	63	0.0476	0.1000	0.0524	52.38 %
$-36.1484 \le X \le 38.5944$	53	63	0.8413	0.8000	0.0413	5.16 %
$X > 38.5944$	7	63	0.1111	0.1000	0.0111	11.11 %
$X > 30$	9	63	0.1429	0.1619	0.0190	11.74 %

C6–C7: the relative frequencies (0.0476, 0.8413, 0.1111) approximate the theoretical probabilities, and the sample mean and SD (1.22 and 29.16) are the ones that define the model.

2.5 AI-use register

Tool: Claude (Anthropic), at claude.ai. **How it helped:** it read the guide, located the official source, processed the CSV that I downloaded (differences, $\bar{x}$, s^2, s, quantiles, counts, table and chart) and drafted this report in LaTeX. It did not invent observations: the data are those of the official CSV. **Estimation method:** $\hat{\mu} = \bar{x}$ and $\hat{\sigma} = s$ (as suggested by the guide). I checked the results against the guide's formulas
Main prompt (originally in Spanish): "Necesito que leas el siguiente PDF [the C10 guide] y haz todo lo que te pide ese documento (...) cumple todo lo que dice la guía" ("I need you to read the following PDF and do everything it asks (...) follow everything the guide says"); afterwards, the TRM CSV downloaded from datos.gov.co.
Link to the conversation: Conversation with Claude.

3 Conclusion and references

Answer. According to the fitted normal model, the probability that the TRM rises by more than 30 COP/USD between two consecutive certifications is 0.1619 (about 16 %); in the data it happened 9 times out of 63 (14.3 %).

Largest error. The event with the largest absolute error is $X < q_{0.10}$ ($EA = 0.0524$): there were only 3 observations versus about 6.3 expected. A plausible reason is model shape: the data are positively skewed, with a longer right tail, so the real left tail is thinner than the normal one; also, with $n = 63$ sampling variation is large (3 observations in a 10 % tail is only a few cases).

Limitations. Fitting and checking use the same data, so this is a descriptive assessment, not proof that the model predicts new changes accurately. The data cover only three months, some changes span weekends, and the dollar's volatility can change with news or policy decisions.

References. Superintendencia Financiera de Colombia (2026). *Tasa de Cambio Representativa del Mercado – TRM* [dataset]. Portal Nacional de Datos Abiertos. https://www.datos.gov.co/d/32sa-8pi3. Updated 11/05/2026; accessed 11/05/2026. Anthropic (2026). *Claude* [AI assistant]. https://claude.ai.